

REQUEST FOR INFORMATION 70RSAT26RFI000016

NOTE: THIS IS A REQUEST FOR INFORMATION (RFI) ONLY.

This RFI is issued solely for information and planning purposes – it does not constitute a Request for Proposal (RFP) or a promise to issue an RFP in the future. This request for information does not commit the Government to contract for any supply or service whatsoever. Further, the Government is not at this time seeking proposals and will not accept unsolicited proposals. Respondents are advised that the U.S. Government will not pay for any information or administrative costs incurred in response to this RFI; all costs associated with responding to this RFI will be solely at the interested party's expense. Not responding to this RFI does not preclude participation in any future RFP, if any is issued.

The Government's explicit intent, in this request for information, is to not receive from respondents any proprietary data, trade secrets, business sensitive information, or information considered CONFIDENTIAL under 18 U.S.C. §1905. The Government's constraint does not in any way relieve contractors from their responsibility to properly mark proprietary data when it is provided, nor does it alleviate any requirement for the Government to protect marked data. The Government is not obligated to protect unmarked data. The information provided in this RFI is subject to change and is not binding on the Government. All submissions become the property of the Federal Government and will not be returned.

The Department of Homeland Security's Science and Technology Directory (DHS/S&T) welcomes all businesses to provide a written response to the following questions, demonstrating capability to fulfill the tasks outlined in the attached draft Statement of Work. If you do not have the capability to fulfill all tasks identified, please inform us of your proposed teaming arrangement that would satisfy the government's full requirements.

Please inform us of your business size and status. If you are a small business, please inform us of your socioeconomic status. If applicable, please inform us of your teaming partner's size, status, and socio-economic status.

THE PURPOSE OF THIS NOTICE IS TO CONDUCT MARKET RESEARCH: The Department of Homeland Security (DHS), Directorate of Science and Technology (S&T) Office Science and Engineering (OSE) is conducting market research to determine technical capabilities to transfer precise time over long-haul fiber that is reproducible, scalable, sustainable for use by end-users, resilient to Positioning, Navigation, and Timing (PNT) threats, independent of the Global Positioning System (GPS), and traceable back to the National Standards Institute and Technology. The resulting product(s) should be a reproducible and enduring pilot capability that can be replicated and scaled and should include:

- Master Timing Node Network
 - Accuracy: 100 nanoseconds (minimal threshold or better)
 - Holdover (Primary Node): 30 days
 - Holdover (Secondary Node): 1 day (Threshold) / 5 days (Objective)
- "Last Mile" Timing Distribution to End-User
 - Accuracy Threshold: 1 microsecond
 - Accuracy Objective: 500 nanoseconds or better
- Timing at both levels must be traceable to UTC(NIST) master clock
- Long-haul fiber means greater than 800 miles, 1,600 miles "round trip"
 - Route includes Local Access And Transport Area (LATA) boundaries

- Architecture that illustrates scalability, sustainability, and repeatability
- Applicability to Power Grid: Western Interconnection, Eastern Interconnection, and Texas Interconnection
- Applicability to the telecommunications service providers at the national level
 - May include National Exchange Carriers Association (NECA) telecommunications service providers

INFORMATION REQUESTED

S&T OSE welcomes all businesses to provide a written response to the questions, demonstrating capability to fulfill the tasks outlined in the attached draft Statement of Work. If you do not have the capability to fulfill all tasks identified, please include a proposed teaming arrangement that would satisfy the Government's full requirements.

- 1) Company Information: Name, phone number, SAM UEI, CAGE Code, POC information, website, etc.
- 2) Available Ordering Vehicles(s): Identify any ordering vehicle(s), including contract numbers, your firm has with the Federal government to include the name of the agency that awarded the vehicle(s).
- 3) A description of any teaming/subcontracting your company anticipates utilizing in support of this requirement.
- 4) Self-Assessment of capabilities experience (considering any teaming arrangements) in performing the following tasks outlined in the DRAFT statement of work provided:

CAPABILITY AREAS	RESPONSE		
	YES	NO	BRIEF COMMENTS (100 words or less)
Extensive experience in fiber communications and networks regarding transfer of time and use within the telecommunications and power industries			
Knowledge of DHS Resilient PNT Reference Architecture and Conformance Framework			
Knowledge of DHS PNT Best Practices and GPS Best Practices			
Experience working with telecommunication critical infrastructure sector and reliance on time			
Knowledge of national power grid and SCADA systems and these systems reliance on precise time			

CAPABILITY AREAS	RESPONSE		
	YES	NO	BRIEF COMMENTS (100 words or less)
Knowledge of DHS Customs and Border Protection (CBP) communication, network and sensors along the southwest border as related to interoperability and compatibility			
Development and transition of knowledge products including reports, testing documentation, guidance documents, methodologies, models, and processes as it relates to critical infrastructure dependency and interdependency on precise time			

- 5) Demonstrated ability to conduct independent assessment of timing data without bias towards a particular approach, technology, other infrastructure equipment, or vendor. Communications and network data assessments are acceptable.
- 6) Describe experience and success utilizing both technically and programmatically utilizing fiber as the transport medium for precise time.
- 7) Demonstrated ability to conduct cyber risk assessment within a fiber network, for example (not inclusive), cable route, distribution nodes, tapping.
- 8) The vendor shall provide a rough staffing plan and skill set to include years of experience in communications, network, systems engineering, technology integration, working with power and telecommunications critical infrastructure sectors.
- 9) Provide S&T with any questions or feedback on the attached SOW.

The total submission page count must not exceed 10 pages and must be legible when viewed at 100% magnification on a computer screen in the application native to the submission's file type (e.g. (*.pdf) Adobe Acrobat or (*.doc/x) Microsoft Word, a cover page may be included and is excluded from the page count). Only one submission will be reviewed per vendor for this RFI. Only electronic submissions will be accepted. Proprietary information, if any, should be minimized and **MUST BE CLEARLY MARKED**. Emails with attachments larger than 50 MB must be broken down into multiple emails with no one attachment exceeding 50 MB. Each electronic submission shall include:

- RFI number 70RSAT26RFI000016 and "Office of Science & Engineering PNT Support" in the Subject Line
- If the submission includes more than one electronic email, please include the file number and total files being submitted (e.g., File 1 of 4, File 2 of 4, etc.) in the Subject Line
- Responses must have a cover letter that includes the following information:

- Company Name
- Company Address
- Point of Contact, phone number, and email address of the point of contact regarding any correspondence between the Government and the vendor
- SAM Unique Entity Identifier (UEI) Number
- Company's Size and Socio-Economic Status information for NAICS code 541330
 - If applicable, teaming partner's Size and Socio-Economic Status
- Technical Point of Contact(s) (Name, title, email, and phone number)
- Contracting Point of Contact(s) (Name, title, email, and phone number)

SUBMISSION INSTRUCTIONS

All Responses shall be submitted to Jennifer.Koons@hq.dhs.gov and John.Whipple@hq.dhs.gov by or before 12:00 PM Eastern Daylight Saving Time (EDT), on Friday, May 1, 2026.

The Government reserves the right to hold one-on-one meetings as a result of responses received from this RFI as part of its market research. Meetings may be held in particular with vendors who provide comprehensive responses to the questions posed.

DRAFT STATEMENT OF WORK

The objective of this effort is to develop and demonstrate a pilot that is reproducible, scalable, and sustainable for use by end-users, and is resilient to Positioning, Navigation, and Timing (PNT) threats. The resulting products from this work shall be a reproducible and enduring pilot capability that can be replicated and scaled in other states.

Positioning, Navigation, and Timing (PNT) services have become an invisible but essential utility for many critical infrastructure operations. Executive Order 13905, “Strengthening National Resilience Through Responsible Use of Positioning, Navigation, and Timing Services,” indicates that the “the disruption or manipulation of these services has the potential to adversely affect the national and economic security of the United States.”

Precision timing from PNT services are particularly essential to many critical infrastructure sectors, and this precision timing is predominantly provisioned through the Global Positioning System (GPS) and other Global Navigation Satellite Systems (GNSS). However, GPS and GNSS are vulnerable to intentional and unintentional interference, and the frequency of some incidents has become increasingly frequent.

To address this strategic risk to U.S. critical infrastructure, this project will pilot an effort to (1) establish a regional network of resilient timing nodes and (2) a means of distributing precision timing to critical infrastructure operators in the local area. The products of this effort shall be scalable and reproducible to allow other states to replicate a similar capability to enable resilient operations for their critical infrastructure.

This effort shall leverage other existing commercial timing standards, government publications, and resources on PNT resilience where applicable, such as the DHS Resilient PNT Reference Architecture, DHS Resilient PNT Conformance Framework, DHS PNT Best Practices, and the NIST PNT Profiles.

The following documents may be helpful to the Contractor in performing the work:

- DHS Resilient PNT Reference Architecture, V1.0
<https://www.dhs.gov/science-and-technology/publication/resilient-pnt-reference-architecture>
- DHS Resilient PNT Conformance Framework (Version 2.0)
<https://www.dhs.gov/publication/st-resilient-pnt-conformance-framework>
- DHS PNT Best Practices (September 2024)
<https://www.dhs.gov/science-and-technology/publication/pnt-best-practices-supporting-critical-infrastructure>
- DHS GPS Best Practices (January 2017)
https://www.cisa.gov/sites/default/files/documents/Improving_the_Operation_and_Development_of_Global_Positioning_System_%28GPS%29_Equipment_Used_by_Critical_Infrastructure_S508C.pdf
- NIST PNT Profiles (Latest Version)
<https://csrc.nist.gov/pubs/ir/8323/r1/final>
- Other Resources on GPS.gov
<https://www.gps.gov/resilience-through-responsible-use-pnt>
- Definition of Critical Infrastructure Sectors
<https://www.cisa.gov/topics/critical-infrastructure-security-and-resilience/critical-infrastructure-sectors>

The Contractor shall develop and demonstrate a pilot of a resilient regional PNT timing network architecture for use by critical infrastructure end-users. This pilot shall consist of (1) a master timing node network; (2) traceability to UTC(NIST); and (3) “last mile” timing distribution to critical infrastructure end-users.

The Contractor shall work with stakeholder partners for all three portions of the pilot. Additionally, the Contractor shall develop documentation and other necessary resources to enable other states to reproduce a similar capability.

Figure 1 shows a conceptual architecture diagram for the pilot effort. The master timing node network consists of Primary Nodes, Secondary Nodes, and Distribution Nodes. The Primary and Secondary nodes shall contain high quality holdover clocks that can continue to operate for a period of time without external timing source input, such as PTP and/or GPS. The Primary Node will have higher quality clocks than the Secondary Node, which also increases its cost. These nodes shall also incorporate concepts from the DHS Resilient PNT Reference Architecture to ensure PNT resilience and treat all external PNT sources as attack surfaces.

The initial pilot may contain a downscaled subset of the timing node network due to funding constraints. However, the pilot must still demonstrate the overall architecture while ensure it is reproducible and provides usable timing information to end-users.

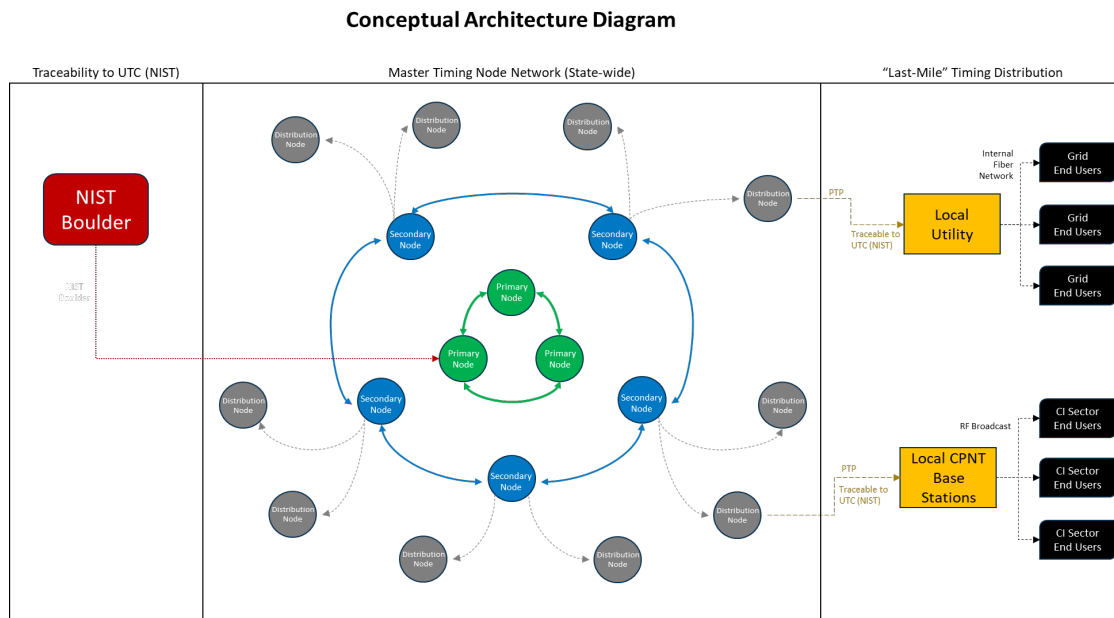


Figure 1. Conceptual Architecture Diagram

Specific Requirements / Tasks

The pilot shall meet the following minimum threshold requirements:

- Master Timing Node Network
 - Accuracy: 100 nanoseconds (or better)
 - Holdover (Primary Node): 30 days
 - Holdover (Secondary Node): 1 day (Threshold) / 5 days (Objective)
- “Last Mile” Timing Distribution to End-User
 - Accuracy Threshold: 1 microsecond
 - Accuracy Objective: 500 nanoseconds or better
- Timing at both levels must be traceable to UTC(NIST)